

# Final project proposal

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## 1 Topic

Conversational system with router-controlled RAG.

## 2 Possible datasets

- MessIRve: <https://huggingface.co/datasets/spanish-ir/messirve>
- Multi-EuP: The Multilingual European Parliament Dataset:  
<https://huggingface.co/datasets/unimelb-nlp/Multi-EuP>
- Spanish Question Answering Corpus (SQAC):  
<https://huggingface.co/datasets/PlanTL-GOB-ES/SQAC>
- MIRACL: <https://huggingface.co/datasets/miracl/miracl-corpus>

## 3 Main objective

Incorporate a retrieval router module that decides whether each query requires external knowledge retrieval before generating a response.

## 4 Secondary objectives

- Build a Spanish conversational system.
- Analyze router behaviour and failure modes.

## 5 Related work

*Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection*  
(Asai et al., 2023).

## 6 Pipeline

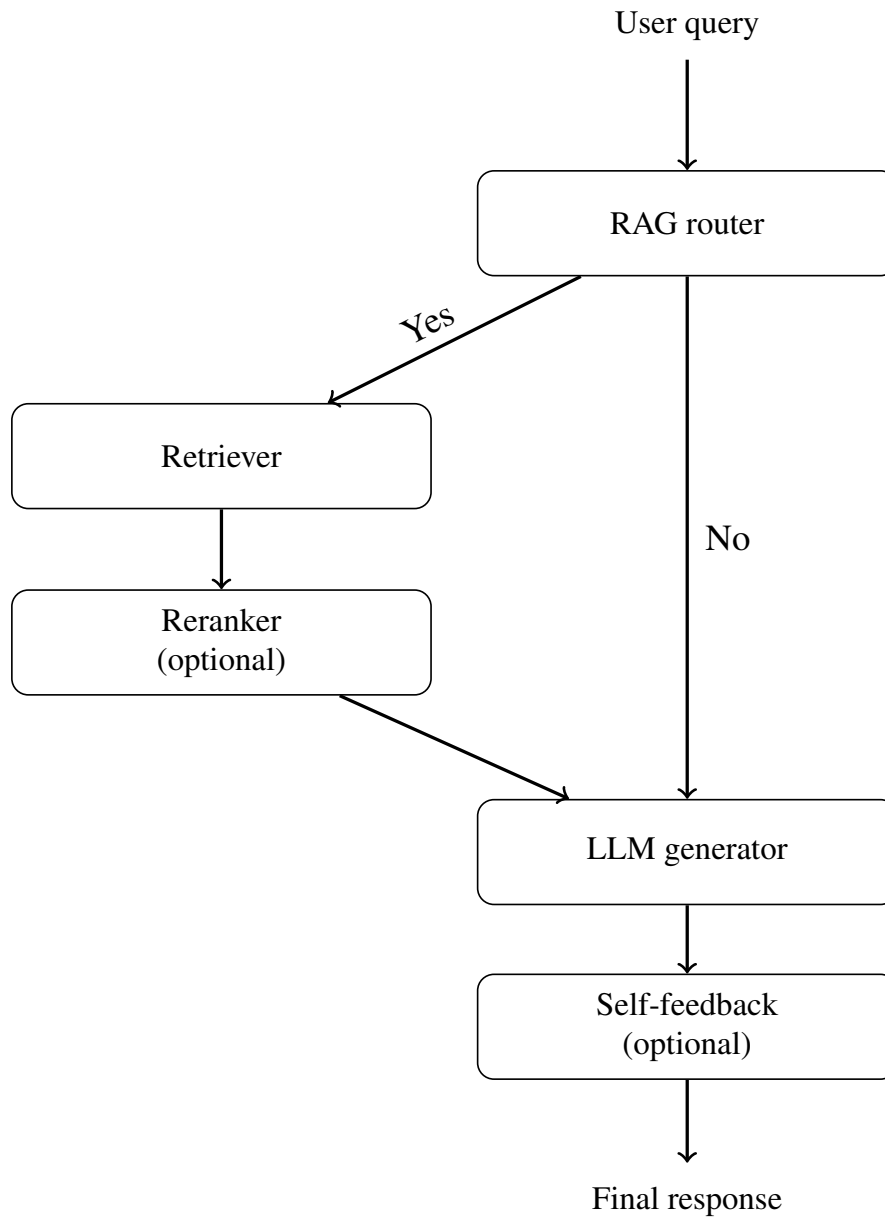


Figure 1: Overview of the pipeline.

## 7 Evaluation plan

### 7.1 Response quality metrics

We will evaluate generated answers against gold references from the datasets using:

- BERTScore (F1): for semantic similarity.
- Token overlap (F1): for lexical similarity.

## 7.2 Retrieval quality metrics

- Recall@k / MRR (when gold retrieval is available, e.g. SQAC / MIRACL datasets).
- Answer quality compared to non-retrieval output (BERTScore + token overlap on gold answers).

## 7.3 Router-specific metrics

- Oracle routing: consider decision as correct if chosen route’s output is closer to gold answer than discarded route’s output (BERTScore + token overlap).
- Informative metric: retrieval rate (% of queries where router chooses “retrieve”).

## 7.4 System-level comparison

To quantify the contribution of the router, we will compare full pipeline configurations in an ablation setting:

- Never RAG
- Always RAG
- Router-controlled RAG

## References

Akari Asai, Zeqiu Wu, Yizhong Wang, Avi Sil, and Hannaneh Hajishirzi. 2023. [Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection](#). In *Proceedings of the Twelfth International Conference on Learning Representations (ICLR)*.